

23CSE301 — Lab Session 07: Decision Trees on Project Data

Project: Audelle — AI-powered audio editor. Dataset: GTZAN subset (24 clips, genres: disco, hiphop, metal, classical), same Hugging Face mirror as `scripts/train_genre_classifier.py`. Features per clip (librosa): spectral centroid mean, ZCR mean, RMS mean, spectral-flatness mean, tempo. Target: genre label.

1. Introduction (project understanding)

Audelle edits audio from natural-language prompts. Before any edit, supervised classifiers read the file: a RandomForest groove classifier (15 EDM feels), a RandomForest genre classifier (10 GTZAN genres), and a RandomForest noise-condition classifier plus SNR regressor. This lab studies their foundation — the decision tree — on our own genre data.

2. Literature survey

scikit-learn `plot_tree` docs (entropy/Gini criteria), the DataCamp decision-tree tutorial (classification pipeline, boundaries), the sklearn iris example (2-D boundary template), and GTZAN spectral genre classification as the MIR benchmark behind our genre head.

3. Methodology (A1–A8)

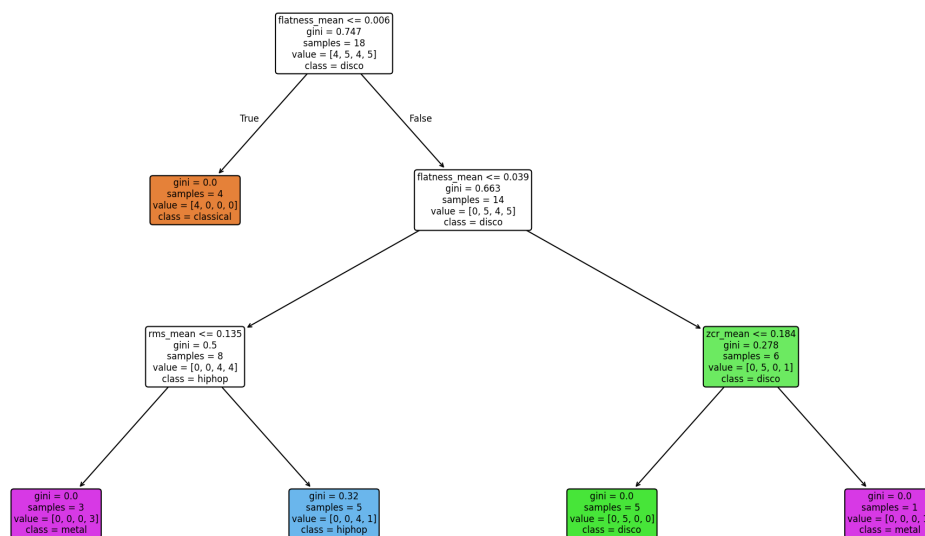
A1 entropy $H = -\sum p \log_2 p$; A2 Gini $= 1 - \sum p^2$; A3 root by information gain on binned features; A4 `bin_feature` with `binning_type/n_bins` defaults; A5 hand-rolled recursive multi-way tree (`max_depth=3`); A6 `plot_tree` figure; A7 centroid-vs-ZCR meshgrid boundary; A8 `GridSearchCV` over `criterion/depth/split`, `cv=3`.

4. Results

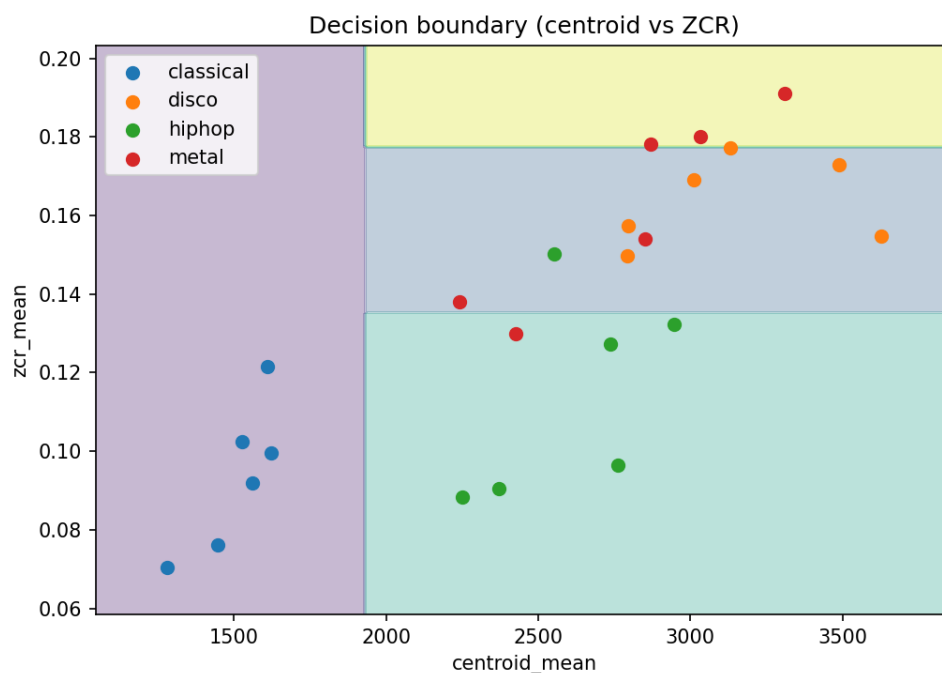
Entropy: 2.0 bits; Gini: 0.75; root: `centroid_mean`; hand-rolled test accuracy: 0.5; tuned: `{'criterion': 'gini', 'max_depth': 3, 'min_samples_split': 2}` (CV 0.75).

Feature	Information gain
<code>centroid_mean</code>	0.9396
<code>rms_mean</code>	0.7859
<code>zcr_mean</code>	0.7728
<code>flatness_mean</code>	0.6609
<code>tempo</code>	0.2387

Decision tree (sklearn, `max_depth=3`):



Decision boundary (centroid vs ZCR):



5. Observations, inferences, discussion

Centroid dominates the root split (gain 0.9396); tempo is near-useless (gain 0.2387) as genres overlap in BPM. The hand-rolled tree trails sklearn: multi-way binned splits fragment 24 samples while binary splits with min_samples_split regularize — why the product uses forests. Rectangular boundary regions are the tree signature. Shallow depth-3 wins tuning: capacity control beats criterion choice here.

6. Conclusion

Balanced 4-class problem quantified, centroid root identified, hand-built + visualized + boundary + tuned pipeline completed on project GTZAN data — directly justifying the product's RandomForest heads as ensembles of these tuned trees.